Diverging Lens: Worked Solutions

Leaving Certificate Physics: Light and Optics

Lens formula

$$1/f = 1/u + 1/v$$

Sign convention: real-is-positive. A real object has u positive. A virtual image has v negative. A diverging (concave) lens has f negative.

The five steps: (1) list the data with signs, (2) write the formula, (3) substitute, (4) calculate, (5) state the answer with units and check it.

Question 1: Find the focal length

An object is 50 cm from a diverging lens and a virtual image is formed 25 cm from the lens. Calculate the focal length of the lens.

Step 1: Data

- u = +50 cm (real object)
- v = −25 cm (virtual image)
- f = ?

Step 2: Formula

$$1/f = 1/u + 1/v$$

Step 3: Substitute

$$1/f = 1/50 + 1/(-25)$$

Step 4: Calculate

$$1/f = 1/50 - 2/50 = (-1)/50$$

$$f = -50 \text{ cm}$$

Step 5: Answer and check

The focal length is −50 cm.

The sign is negative, as it must be for a diverging lens. The image is closer to the lens than the object, which is what a diverging lens produces.

Question 2: Find the image distance

A concave lens has a focal length of 30 cm. An object is placed 60 cm from the lens. Calculate the distance of the image from the lens.

Step 1: Data

- f = −30 cm (diverging)
- u = +60 cm
- v = ?

Step 2: Formula

$$1/f = 1/u + 1/v$$

Rearrange for v:

$$1/v = 1/f - 1/u$$

Step 3: Substitute

$$1/v = 1/(-30) - 1/60$$

Step 4: Calculate

$$1/v = (-2)/60 - 1/60 = (-3)/60 = (-1)/20$$

$$v = -20 \text{ cm}$$

Step 5: Answer and check

The image is 20 cm from the lens.

The negative sign shows the image is virtual and on the same side of the lens as the object. A diverging lens always gives a virtual, upright, diminished image.

Question 3: Find the image distance

A diverging lens has a focal length of 15 cm. An object is placed 45 cm in front of it. How far from the lens is the image formed?

Step 1: Data

- f = −15 cm
- u = +45 cm
- v = ?

Step 2: Formula

$1/v = 1/f - 1/u$

Step 3: Substitute

$1/v = 1/(-15) - 1/45$

Step 4: Calculate

$1/v = (-3)/45 - 1/45 = (-4)/45$

$v = (-45)/4 = -11.25$ cm

Step 5: Answer and check

The image is 11.25 cm from the lens.

It is virtual (negative sign) and on the same side as the object. It lies closer to the lens than the focal length (15 cm), which is always true for a diverging lens.

Question 4: Find the object distance

A concave lens has a focal length of 20 cm. A virtual image is formed 8 cm from the lens. Calculate the distance of the object from the lens.

Step 1: Data

- f = −20 cm
- v = −8 cm (virtual image)
- u = ?

Step 2: Formula

$1/f = 1/u + 1/v$

Rearrange for u:

$1/u = 1/f - 1/v$

Step 3: Substitute

$1/u = 1/(-20) - 1/(-8)$

$1/u = (-1)/20 + 1/8$

Step 4: Calculate

$1/u = (-2)/40 + 5/40 = 3/40$

$u = 40/3 = 13.3$ cm

Step 5: Answer and check

The object is 13.3 cm from the lens.

The positive sign shows the object is real. It is further from the lens than the image (8 cm), as expected for a diverging lens.

Question 5: Find the focal length

An object is placed 100 cm from a concave lens and the image is formed 20 cm from the lens. Calculate the focal length of the lens.

Step 1: Data

- u = +100 cm
- v = −20 cm
- f = ?

A diverging lens only forms virtual images of real objects, so the image distance must be taken as negative.

Step 2: Formula

$$1/f = 1/u + 1/v$$

Step 3: Substitute

$$1/f = 1/100 + 1/(-20)$$

Step 4: Calculate

$$1/f = 1/100 - 5/100 = (-4)/100 = (-1)/25$$

$$f = -25 \text{ cm}$$

Step 5: Answer and check

The focal length is −25 cm.

The negative sign confirms a diverging lens. The image (20 cm) is closer to the lens than the object (100 cm), as expected.

Summary of answers

1. f = −50 cm
2. v = −20 cm (image 20 cm from lens, virtual)
3. v = −11.25 cm (image 11.25 cm from lens, virtual)
4. u = 13.3 cm
5. f = −25 cm